

ASSIGNMENT No: 2

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TITLE: Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Problem Statement: Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives: To understand and implement default, parameterized, and copy constructors for object initialization.

Theory: Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

Exercise

1. Create a Student class using default and parameterized constructors to initialize student name and roll number.

Code:

```
J assignment2.java > Student
1 //Assignment 2
2 class Student{
3     String name;
4     int age;
5     public String id;
6
7     //Default constructor-if not written no error is returned
8     Student(){
9         name = "Bob";
10        age = 95;
11    }
12
13    //Parameterized constructor
14    Student(String n, int a){
15        name = n;
16        age = a;
17    }
18
19    //Copy Constructor
20    Student(Student s){
21        name = s.name;
22        age = s.age;
23    }
24
25    void display(){
26        System.out.println("Name: " + name);
27        System.out.println("Age: " + age);
28    }
29 }
30 public class assignment2{
31     Run | Debug
32     public static void main(String[] args){
33         Student student1 = new Student();
34         student1.display();
35     }
36 }
```

Output:

```
PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS
[Running] cd "c:\Users\arhaa\OneDrive\Desktop\java\" && javac assignment2.java && java assignment2
Name: Bob
Age: 95
[Done] exited with code=0 in 0.922 seconds
```

2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

Code:

```
J mobileInventory.java > ...
1  class Mobile{
2      String name;
3      int price;
4
5      //Default constructor-if not written no error is returned
6      Mobile(){
7          name = "Samsung S26 Ultra";
8          price = 120000;
9      }
10
11     //Parameterized constructor
12     Mobile(String n, int p){
13         name = n;
14         price = p;
15     }
16
17     //Copy Constructor
18     Mobile(Mobile m){
19         name = m.name;
20         price = m.price;
21     }
22
23     void display(){
24         System.out.println("Name: " + name);
25         System.out.println("Price: " + price);
26     }
27 }
28 public class mobileInventory{
29     Run | Debug
30     public static void main(String[] args){
31         Mobile mobile1 = new Mobile();
32         mobile1.display();
33     }
34 }
```

Outcome:

PROBLEMS 6

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\arhaa\OneDrive\Desktop\java> & 'C:\Program Files\Java\jdk-26.0.
workspaceStorage\02e7f2cfce92a0ba8e30606ade6b2c24\redhat.java\jdt_ws\java_8f
Name: Samsung S26 Ultra
Price: 120000
PS C:\Users\arhaa\OneDrive\Desktop\java>
```